Style Guide

1. Character Spacing

- Add a space before and after operators
- No spaces before or after array operators
- No space between a function name and its parameters

2. Blank Lines

Add a blank line:

- Between functions
- Between conditional statements
- Before and after comments
- Between the file header and the rest of the file
- Between control structures

3. Indentation

- Use 4 spaces per indent
- Indent in for loops and while loops even if not required in the coding language
- Indent conditionals
- Indent within function declarations

4. Variable Declaration Placement and Alignment

- Using C++11 auto type deduction for longer types
- Also using templates for repeated use of functions with different types
- One variable declaration per line
- Declare variables at the beginning of a function

5. Case Statement Alignment

- Follow the rules in **Section 3**
- Opening braces go inline with the control structure declaration

6. Comment Layout

- Place comments before what they describe, or inline if short enough
- Place function explanations before the function
- Put comments at the top of each file explaining its purpose
- Include a README that describes the purpose of each file

7. Additional Attributes

- **Constants:** all caps with underscores
- **Variables:** capitalized words separated by underscores
- **File names:** all lowercase
- **Classes:** same as variables
- **Functions:** same as variables

8. Style Guide Enforcement

We will use a linter to strictly enforce these guidelines, with no exceptions. Anyone who doesn't follow the rules will be temporarily suspended from committing new code.